CLAPS: A Multi-LLM Framework for Predicting Social Anxiety via Cognitive-Behavioral Theory of Social Anxiety in VR

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Project

Motivations

Need for Cognitive Symptom Modeling

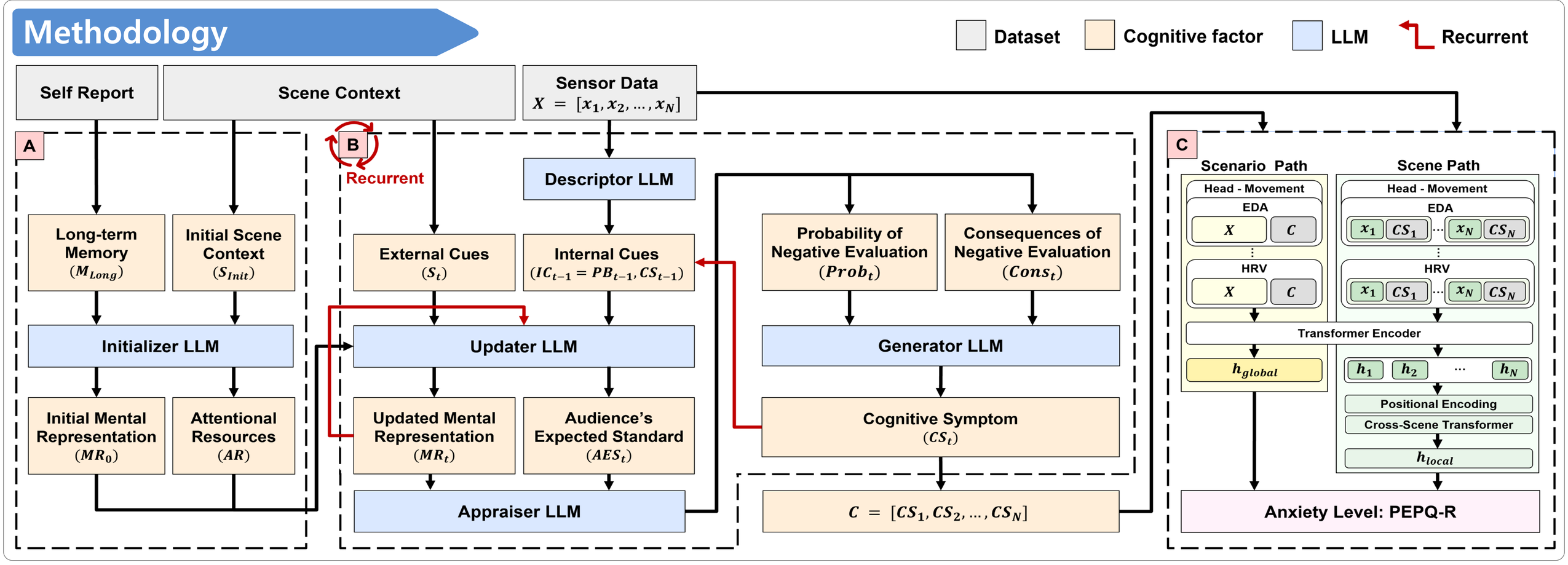
- Social anxiety **involves physiological, behavioral, and cognitive symptoms**, but existing models mainly focus on the first two.
- Cognitive symptoms are **difficult to observe or measure.** So, we should use **theory-based LLMs** to generating them.

Need for Multimodal Alignment

- Physiological, behavioral symptoms are captured as **time series**, while cognitive symptoms are generated in **natural language.**
- These **heterogeneous data need to be aligned** within the same scene for anxiety prediction.

We propose CLAPS, a framework generates cognitive symptoms grounded in the cognitive-behavioral theory of social anxiety and aligns them with co-occurring physiological and behavioral data for anxiety prediction.

Methodology



Experiment

Results averaged over 8-fold cross-validation. Two-sided paired Wilcoxon signed-rank test across 64 test samples.
$*$ $p < 0.05$; $**$ $p < 0.01$; $***$ $p < 0.001$. Lower MAE and RMSE indicate better performance.

Dataset

- 32 participants
- 2 VR scenarios
  - Self-introduction
  - Job interview
- 30 time-series features
  - Physiological
  - head movement
  - eye tracking
- Prediction
  - Nine PEPQ-R items

Performance Comparison

Model	MAE ↓	RMSE ↓
Random Forest	1.745 ± (0.278)***	2.300 ± (0.270)***
XGBoost	1.703 ± (0.201)***	2.235 ± (0.221)***
LightGBM	1.731 ± (0.235)***	2.228 ± (0.242)***
SAFER	1.459 ± (0.271)**	1.769 ± (0.274)**
DepGuard	1.384 ± (0.218)	1.728 ± (0.227)**
Health-LLM	1.564 ± (0.327)***	1.995 ± (0.336)***
CLAPS (ours)	**1.268 ± (0.180)**	**1.558 ± (0.183)**

CLAPS outperformed all evaluated baselines

- Combining cognitive modeling and multimodal alignment supports anxiety prediction.

Ablation Study

Model	MAE ↓	RMSE ↓
CLAPS w/o Cognitive Symptom	1.369 ± (0.223)	1.664 ± (0.271)
CLAPS w/o Scene Path	1.344 ± (0.172)**	1.619 ± (0.201)**
CLAPS (ours)	**1.268 ± (0.180)**	**1.558 ± (0.183)**

- Cognitive symptoms
  - **Provide additional context** for anxiety prediction through LLM-generated cognitive symptoms.
- Scene-level alignment
  - Supports both overall scenario patterns and **scene-level temporal dependencies.**

Ongoing research

CLAPS limitation

1. Depends on predefined scene boundaries
2. Pre-session reports and manual scene context
3. Cognitive model specialized for social anxiety

How we addresses it

- **A** : Segment VR video into 10–30s windows, informed by Newell's cognitive band theory
- **B** : Replace pre-session reports with sensor data and manual scene context with VLM-extracted context
- **C** : Use ACT-R, a general theory of cognition

